

SYDNEY TECHNICAL HIGH SCHOOL

(Est 1911 - celebrating 50 years at Bexley)



Mathematics

YEAR 11 YEARLY EXAMINATION PRELIMINARY HSC ASSESSMENT TASK 3 SEPTEMBER 2006

General Instructions

- Working time allowed – 120 minutes.
- Write using black or blue pen.
- Approved calculators may be used.
- All necessary working should be shown.
- Start each question on a new page.
- Attempt all questions.
- Questions are of equal value.
- Full marks may not be awarded if working is poorly set out or difficult to read.

NAME : _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	TOTAL

Question 2 (13 marks) (Start a new page)**Marks**

a) Evaluate $\sqrt{\frac{284.6}{8.3 \times 6.2}}$ correct to 2 significant figures. 2

b) Solve $\tan \theta = 0.3$ for $0 \leq \theta \leq 360^\circ$ 2
giving answers correct to the nearest degree

c) The price of an item increases from \$1.50 to \$3.60. 2
What percentage increase is this ?

d) Solve $|x - 3| \leq 10$ 2

e) State the domain of the function $y = 4\sqrt{x - 5}$ 1

f) If α and β are the roots of the equation $x^2 - 2x - 7 = 0$

find the value of i) $\alpha + \beta$ 1

ii) $\alpha \times \beta$ 1

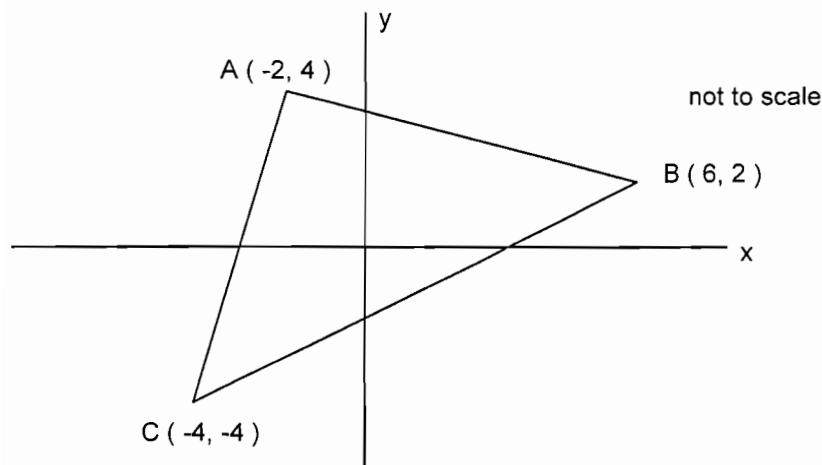
iii) $\alpha^3\beta + \alpha\beta^3$ 2

Question 3 (13 marks) (Start a new page)**Marks**

a) If $\frac{2}{2-\sqrt{3}} = a + \sqrt{b}$ find the value of a and b given they are both rational. 3

b) Find the perpendicular distance from the point $(2,3)$ to the line $3x + y + 2 = 0$. 1

c) The diagram below shows the points $A(-2,4)$, $B(6,2)$ and $C(-4,-4)$



i) Calculate the length of the interval BC. 1

ii) Find the gradient of the line BC. 1

iii) Find the coordinates of M, the midpoint of BC. 1

iv) Show that the equation of l , the perpendicular bisector of BC is $5x + 3y - 2 = 0$. 2

v) Show that l passes through A. 1

vi) What does the result in part v) tell us about triangle ABC. 1

vii) Hence or otherwise find the area of triangle ABC. 2

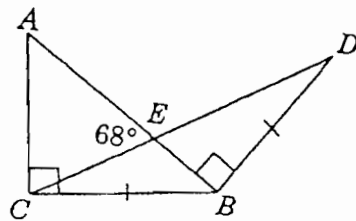
Question 4 (13 marks) (Start a new page)**Marks**a) Find the minimum value of $x^2 - 4x + 14$

2

b) On a number plane sketch the region described by $y \geq 4x - x^2$

2

c)



ABC is a right angled triangle in which $\angle ACB = 90^\circ$.
Triangle CDB is isosceles, in which $CB = DB$.
 $\angle AEC = 68^\circ$ and $\angle EBD = 90^\circ$.
Find $\angle DCB$, giving reasons.

2

d) Differentiate the following

i) $\frac{5}{x^2}$

1

ii) $\sqrt{2x+7}$

2

iii) $\frac{2x-1}{x+1}$

2

iv) $3x^2(x-2)^4$

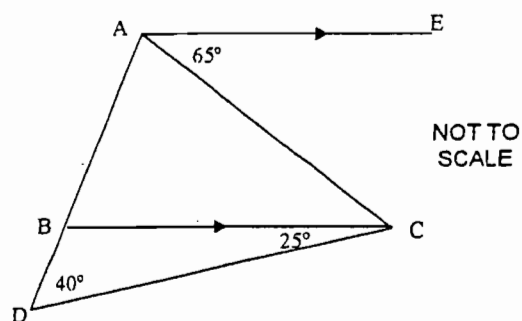
2

Question 5 (13 marks) (Start a new page)**Marks**

a) Evaluate $\lim_{x \rightarrow 4} \frac{x^2 + 2x - 24}{x - 4}$

2

b)



In the diagram above, AE is parallel to BC ,
 $\angle BCD = 25^\circ$, $\angle BDC = 40^\circ$ and $\angle EAC = 65^\circ$.

3

Copy the diagram onto your answer sheet
and show that triangle ABC is isosceles.

c) Find the equation of the normal to $y = x^3 - 6x - 2$

3

at the point $(1, -7)$

d) Simplify $\frac{1 + \cos \theta}{1 - \sin \theta} \div \frac{1 + \sin \theta}{1 - \cos \theta}$

3

e) If $f(x) = x(x+1)(x+2)$

2

find in simplest terms an expression for $f(x+1) - f(x)$

Question 6 (13 marks) (Start a new page)

Marks

a) The quadratic expression $Q(x)$ is given by $Q(x) = x^2 - (2 + k)x + 4$.

i) For what values of k is $x = -2$ a root of $Q(x) = 0$. 2

ii) For what value of k does $Q(x) = 0$ have real roots. 3

b) Find a quadratic equation which has roots $1 + \sqrt{3}$ and $1 - \sqrt{3}$. 2

Express your answer in the form $ax^2 + bx + c = 0$ where a , b and c are real.

c) Find the coordinates of the point on the curve $y = x^2 + 6x + 2$ 3

at which the tangent to the curve is parallel to the line $y = 2x + 3$.

d) A triangle has sides of length 8 cm , 11 cm and 16 cm.

i) Find the size of the largest angle (nearest degree) 2

ii) Find the area of the triangle (correct to 1 decimal place) 1

Question 7 (13 marks) (Start a new page)**Marks**

a) Solve $x^6 + 7x^3 - 8 = 0$

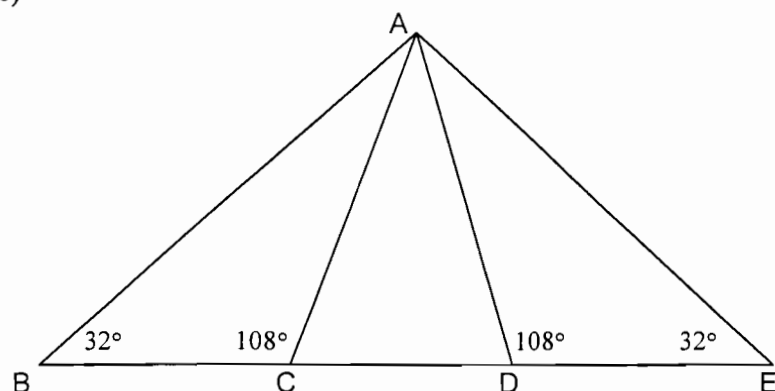
3

b) Find the values of A , B and C if

3

$$3x^2 - 7x + 5 \equiv Ax(x - 1) + Bx + C$$

c)



Given that $BE = 30 \text{ metres}$ find the length of AC
(correct to 1 decimal place)

3

d) The roots of the equation $ax^2 + bx + c = 0$ differ by 4.

4

Show that $b^2 = 4ac + 16a^2$

Teacher's Name: _____

Student's Name/N^o: _____Q1

a. $(x-2)(x-3)=0$

$x=2, 3$

d. $-10 \leq x-3 \leq 10$

$-7 \leq x \leq 13$

b. $\frac{5(2x-5)-2(2x-1)}{10}$

$= \frac{6x-23}{10}$

e. domain $x \geq 5$

f. i) $\alpha + \beta = 2$

c. $15t^2$

ii) $\alpha\beta = -7$

d. $8\sqrt{2} + \sqrt{2}$

$= 9\sqrt{2}$

iii) $\alpha\beta(\alpha^2 + \beta^2)$

$= \alpha\beta[(\alpha+\beta)^2 - 2\alpha\beta]$

e. i) $\frac{2}{\sqrt{3}}$

$= -7[2^2 - 2(-7)]$

ii) 0

$= -126$

f. $y' = 2x - 4$

when $x=1$

$m = -2$

Q3

a. $\frac{2}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}}$

$= \frac{4+2\sqrt{3}}{4-3}$

$= 4+2\sqrt{3}$

$= 4 + \sqrt{12}$

$\therefore a=4 \quad b=12$

g. i) $\Delta = 9 - 8k$

ii) $\Delta < 0 \quad (a > 0)$

$9 - 8k < 0$

$k > \frac{9}{8}$

Q2

b. $\frac{12 \times 3 + 3 \times 1 + 2 \times 1}{\sqrt{3^2 + 1^2}}$

$d = \frac{11}{\sqrt{10}}$

a. 2.4

$= \frac{11}{\sqrt{10}} \text{ units}$

b. $\theta = 17^\circ, 197^\circ$

Teacher's Name:

Student's Name/N^o:

$$c. \quad i) \quad d = \sqrt{(6 - -4)^2 + (2 - -4)^2}$$

$$= \sqrt{136} \text{ units}$$

$$ii) \quad m = \frac{2 - -4}{6 - -4}$$

$$= \frac{6}{10}$$

$$= \frac{3}{5}$$

$$iii) \quad M = (1, -1)$$

$$iv) \quad m_1 = -\frac{5}{3}$$

$$\therefore y + 1 = -\frac{5}{3}(x - 1)$$

$$3y + 3 = -5x + 5$$

$$5x + 3y - 2 = 0$$

$$v) \quad \text{sub. } x = -2, y = 4$$

$$5(-2) + 3(4) - 2 = 0$$

$$\therefore 0 = 0$$

$\therefore A$ lies on l

$$vi) \quad \triangle ABC \text{ is isosceles}$$

(or $AB = AC$)

$$vii) \quad d_{AB} = \sqrt{(-2 - 1)^2 + (4 - -1)^2}$$

$$= \sqrt{9 + 25}$$

$$= \sqrt{34}$$

$$\therefore \text{Area} = \frac{1}{2} \times \sqrt{136} \times \sqrt{34}$$

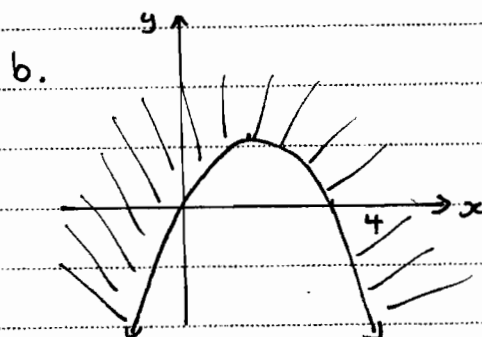
Q4

$$a. \quad x = \frac{4}{2}$$

$$= 2$$

$$\therefore \min = 2^2 - 4 \times 2 + 14$$

$$= 10$$



c.

$$\angle DEB = 68^\circ \text{ (Vertically opposite)}$$

$$\angle EDB = 22^\circ \text{ (angle sum of triangle)}$$

$$\therefore \angle DCB = 22^\circ \text{ (angles opposite equal sides equal)}$$

$$d. \quad i) \quad -10x^{-3}$$

$$ii) \quad (2x + 7)^{-\frac{1}{2}}$$

$$iii) \quad \frac{(x+1)^2 - (2x-1)(1)}{(x+1)^2}$$

$$= \frac{3}{(x+1)^2}$$

$$iv) \quad 6x(x-2)^4 + 3x^2 \cdot 4(x-2)^3$$

$$= 6x(x-2)^4 + 12x^2(x-2)^3$$

Q5

$$a. \lim_{x \rightarrow 4} \frac{(x-4)(x+6)}{x-4}$$

$$= 10$$

$$b. \angle ACB = 65^\circ \text{ (alternate angles, } AE \parallel BC)$$

$$\angle ABC = 65^\circ \text{ (exterior angle equals sum of two remote interior angles)}$$

$\therefore \triangle ABC$ is isosceles (two equal angles)

$$c. y' = 3x^2 - 6$$

when $x = 1$

$$m_T = -3$$

$$\therefore m_N = \frac{1}{3}$$

$$\therefore y + 7 = \frac{1}{3}(x - 1)$$

$$3y + 21 = x - 1$$

$$x - 3y - 22 = 0$$

$$d. \frac{1 + \cos \theta}{1 - \sin \theta} \times \frac{1 - \cos \theta}{1 + \sin \theta}$$

$$= \frac{1 - \cos^2 \theta}{1 - \sin^2 \theta}$$

$$= \frac{\sin^2 \theta}{\cos^2 \theta}$$

$$= \tan^2 \theta$$

$$e. f(x+1) - f(x)$$

$$= (x+1)(x+2)(x+3) - x(x+1)(x+2)$$

Q6

$$a. i) \text{ sub } x = -2$$

$$(-2)^2 - (2+k)(-2) + 4 = 0$$

$$8 + 4 + 2k = 0$$

$$2k = -12$$

$$k = -6$$

$$b. ii) \text{ real roots } \Rightarrow \Delta \geq 0$$

$$(2+k)^2 - 4 \times 4 \geq 0$$

$$k^2 + 4k + 4 - 16 \geq 0$$

$$k^2 + 4k - 12 \geq 0$$

$$(k+6)(k-2) \geq 0$$

$$k \leq -6, k \geq 2$$

$$b. \alpha + \beta = 2$$

$$\alpha \beta = -2$$

$$\therefore x^2 - 2x - 2 = 0$$

$$c. y' = 2x + 6$$

$$m_{\text{line}} = 2$$

$$\therefore 2x + 6 = 2$$

$$2x = -4$$

$$x = -2$$

$$\therefore \text{point } (-2, -6)$$

$$d. i) \cos A = \frac{8^2 + 11^2 - 16^2}{2 \times 8 \times 11}$$

$$\therefore A = 114^\circ$$

Teacher's Name:

Student's Name/N^o:Q7

$$a. (x^3 + 8)(x^3 - 1) = 0$$

$$\alpha = -\frac{1}{2} \left(\frac{b}{a} + 4 \right) \text{ from (1)}$$

sub. into (2)

$$x^3 = -8, \quad x^3 = 1$$

$$x = -2, \quad x = 1$$

$$-\frac{1}{2} \left(\frac{b}{a} + 4 \right) \left(-\frac{1}{2} \left(\frac{b}{a} + 4 \right) + 4 \right) = \frac{c}{a}$$

$$-\frac{1}{2} \left(\frac{b}{a} + 4 \right) \left(\frac{-b}{2a} - 2 + 4 \right) = \frac{c}{a}$$

$$-\frac{1}{2} \left(\frac{b}{a} + 4 \right) \left(\frac{-b}{2a} + 2 \right) = \frac{c}{a}$$

$$-\frac{1}{2} \left(\frac{-b^2}{2a^2} + \frac{2b}{a} - \frac{4b}{2a} + 8 \right) = \frac{c}{a}$$

$$\frac{-b^2}{2a^2} + \frac{2b}{a} - \frac{4b}{2a} + 8 = -\frac{2c}{a}$$

$$-b^2 + 16a^2 = -4ac$$

$$\therefore b^2 = 4ac + 16a^2$$

$$c. \frac{AB}{\sin 32^\circ} = \frac{30}{\sin 116^\circ}$$

$$\therefore AB = 17.6877$$

$$\frac{AC}{\sin 32^\circ} = \frac{AB}{\sin 108^\circ}$$

$$\therefore AC = 9.9 \text{ m}$$

a. let roots equal $\alpha, \alpha + 4$

$$\therefore 2\alpha + 4 = -\frac{b}{a} \quad (1)$$

$$\alpha(\alpha + 4) = \frac{c}{a} \quad (2)$$